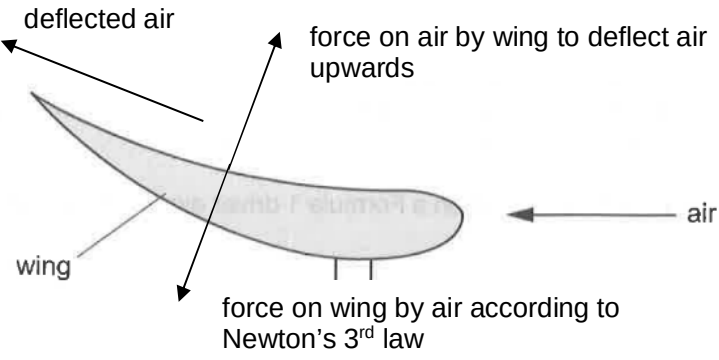


be less than zero and thus negative.

8(a)	Using $v^2 = u^2 + 2as$ $a = \frac{v^2 - u^2}{2s}$ $= \frac{0 - \left(\frac{185 \times 1000}{3600}\right)^2}{2(80)}$ $= -16.5 \text{ m s}^{-2}$ Deceleration = $16.5 \text{ m s}^{-2} = 17 \text{ m s}^{-2}$
8(b)(i)	Using $a = \frac{v^2}{r}$ Maximum lateral acceleration is $4g$. $v^2 = ar = (4g)(30)$ $v = 34 \text{ m s}^{-1}$
8(b)(ii)	From Fig. 8.2, as the tyre is heated up, it becomes softer and thus can deform and fill up the air gaps between the tyre and the road. This increase in contact area will result in an increase in friction, as shown by the increase in the coefficient of friction in Fig. 8.1. The higher friction will enable the car to have a higher acceleration or deceleration. For example, a higher lateral friction will allow the car to go round a corner at higher speed.

No.	Solution
8(c)	 The incoming air was deflected upwards by the top of the wing. The air thus experienced a change in momentum which is upwards that is perpendicular to the wing. According to Newton's 2nd law, a resultant force acts on the air and it is exerted by the wing. According to Newton's 3rd law, an equal and opposite force acts on the wing by the air. This downwards force on the wing appears to increase the weight of the car.
8(d)(i)	For vertical equilibrium, $N_R + N_F = W$ $N_F = W - N_R$ Taking moments about the C.G., clockwise moments = anti-clockwise moments $N_R x_R = N_F x_F + Dh = (W - N_R) x_F + Dh$ $N_R x_R = W x_F - N_R x_F + Dh$ $N_R x_R + N_R x_F = W x_F + Dh$ $N_R = \frac{W x_F + Dh}{x_R + x_F}$
8(d)(ii)	From above, substitute $N_R = W - N_F$ into $N_R = \frac{W x_F + Dh}{x_R + x_F}$ $W - N_F = \frac{W x_F + Dh}{x_R + x_F}$ $N_F = \frac{W(x_R + x_F) - W x_F - Dh}{x_R + x_F} = \frac{W x_R - Dh}{x_R + x_F}$
8(d)(iii)	When the car is accelerating, the driving force D increase. From the equation $N_R = \frac{W x_F + Dh}{x_R + x_F}$, N_R increases. From $N_F = \frac{W x_R - Dh}{x_R + x_F}$, N_F decreases. As such, the front contact force decrease while the rear contact force increases resulting in a weight transfer to the rear from the front.
8(e)(i)	Assuming all loss in K.E. is converted to thermal energy in the 4 disc brakes. From conservation of energy, loss in K.E. = gain in thermal energy

No.	Solution
	$\frac{1}{2}m_{car}v^2 - 0 = m_{4 \text{ brakes}}c\Delta\theta$ $\Delta\theta = \frac{m_{car}v^2}{2m_{4 \text{ brakes}}c} = \frac{(750)(51.39)^2}{2(4 \times 1.2)(1130)} = 182.6 \text{ }^{\circ}\text{C} = 180 \text{ }^{\circ}\text{C}$
8(e)(ii)	Assuming all loss in K.E. is converted to thermal energy in the 4 disc brakes and none to other parts of the car.
8(e)(iii) 1.	The many small holes will increase air flows and thus increase cooling of the brake disc and prevents overheating which will reduce the effectiveness of the brakes.
8(e)(iii) 2.	When the disc is wet due to rain etc, the holes enable the water to be dispelled more easily and prevents the formation of a thin film of water between the brake disc and brake pads. Such a thin film of water will reduce friction and thus braking effectiveness significantly.